Life Insurance Profitability & Cashflow Modelling

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1 Introduction

This project evaluates the expected profitability of a 20-year unit-linked term assurance policy issued to a life aged 58. An Excel model is used to construct an updated multiple decrement table, project policy cashflows and calculate the net present value of profit and profit margin. The effect of surrender benefit payment timing on profitability is also investigated.

1.1 Policy Assumptions

The annual premium is £2,000, payable in advance, with 65% allocated to the unit fund. The fund incurs an annual management charge of 1.1%, deducted at the end of each year before benefits are paid, and a bid-offer spread of 3%. The policy provides the following benefits:

- **Death:** The greater of £7,000 or the bid value of units, payable at the end of the year of death.
- **Redundancy:** $(100 + t/2)\%$ of the bid value of the unit fund, payable at the end of the year of redundancy, where $t = 1, \ldots, 20$.
- **Surrender:** 60% of total premiums paid before surrender, without interest, payable at the end of the year of surrender.

Initial expenses are £800, renewal expenses are 3% of the annual premium (including at policy inception), and terminal expenses are 2% of the benefit payable. The expected annual return on units is 2.5% for the first 10 years, increasing by 2% of the previous year's return each year thereafter. The cash account earns 2% per annum, and profits are valued using a risk discount rate of 5%.

1.2 Multiple Decrement Assumptions

The policy is subject to three independent decrements: death, redundancy and surrender. The underlying rates of decrement are assumed constant within each year of age. The original forces of mortality and surrender are adjusted by -8% and $+5\%$, respectively, while redundancy rates remain unchanged.

1.3 Surrender Timing

The baseline policy pays surrender benefits at the end of the year of surrender. Alternative policies are considered where benefits are paid at the end of the $1/m$th of the year of surrender, for $m = 2, 4, 8, 16, 32, 64$, and immediately upon surrender. For these alternatives, dependent surrender is assumed to occur uniformly throughout each year of age. All other policy assumptions remain unchanged.

2 Adjusting MDT

The data we receive is a multiple decrement table, listing the quantities $(aq)_x^d, (aq)_x^r$ and $(aq)_x^s$. This refers to the dependent probabilities that a life aged x, will leave the active state due to death, redundancy or surrender respectively, in the next year. We are told that there have been some changes to these values, namely that the death rates μ_x^d have decreased by 8% on every year of age, furthermore the surrender rates μ_x^s have increased by 5% by each year of age and the redundancy rates remain unchanged. Denote the old death, redundancy and surrender rates by $\hat{\mu}_x^d,\ \hat{\mu}_x^r,\ \hat{\mu}_x^s$. To obtain our new probabilities, we can use the equation:

$$\hat{\mu}_x^{r_i} = -\frac{(aq)_x^{r_i}}{\sum_{j=1}^{3}(aq)_x^{r_j}} \cdot \ln\left(1 - \sum_{j=1}^{3}(aq)_x^{r_i}\right),$$

where r_1, r_2, r_3 refer to decrements d, r, s respectively.

Age	$(aq)_x^d$	$(aq)_x^r$	$(aq)_x^s$	$(aq)_x$
58	0.000708	0.005278	0.073208	0.079193
59	0.000771	0.006313	0.073190	0.080274
60	0.000842	0.007865	0.064520	0.073227
61	0.000921	0.013451	0.056760	0.071132
62	0.001010	0.016214	0.054672	0.071897
63	0.001110	0.019961	0.053739	0.074811
64	0.001223	0.031948	0.052566	0.085737
65	0.001350	0.032428	0.051896	0.085674
66	0.001492	0.035122	0.050942	0.087556
67	0.001652	0.037130	0.050835	0.089617
68	0.001831	0.042205	0.050472	0.094508
69	0.002034	0.048233	0.038839	0.089105
70	0.002261	0.045526	0.037642	0.085428
71	0.002516	0.032261	0.037158	0.071935
72	0.002802	0.025290	0.035734	0.063826
73	0.003124	0.021996	0.034975	0.060095
74	0.003486	0.021473	0.035060	0.060019
75	0.003893	0.010054	0.029685	0.043632
76	0.004351	0.003249	0.015178	0.022778
77	0.004866	0.001283	0.000000	0.006149

Table 1: New multiple decrement table.

We can then apply our rate differences by calculating $\mu_x^d = 0.92\hat{\mu}_x^d,\ \mu_x^r = \hat{\mu}_x^r$ and $\mu_x^s = 1.05\hat{\mu}_x^s$ for all $x \in \{58, 59, 60, \ldots, 77\}$. From these we can recalculate our values for $(aq)_x^{r_i}$ by calculating:

$$(aq)_x^{r_i} = \frac{\mu_x^{r_i}}{\sum_{j=1}^{3}\mu_x^{r_j}} \cdot \left(1 - e^{-\sum_{j=1}^{3}\mu_x^{r_j}}\right), \quad \forall i \in \{1,2,3\},\ \forall x \in \{58, 59, \ldots, 77\}.$$

This results of the new multiple decrement table are shown in table 1.

3 Projecting Unit Linked Cashflows

Now we have adjusted our multiple decrement table taking into account our new rates, we can begin to project our unit linked cashflow. We begin with projecting the cashflow in the unit fund account. To do this we need to take into account our annual management charge (AMC) at a rate of 1.1% of the gross year end (GYE) price, the unit fund allocation (65%) as well as the bid/offer (B/O) spread (Δ) at a rate of 3%. We also have an expected return in the unit fund account (ρ_k) defined in year k as:

$$\rho_k = \begin{cases} 0.025, & \text{if } k \leq 10, \\ 0.025 \cdot (1.02)^{(k-10)}, & \text{if } k \geq 11. \end{cases}$$

Within this projection we have the following values, note that below P refers to the annual premium:

- Year: $k \in \{1, 2, \ldots, 20\}$.
- Initial Balance: $IB_k = NYE_{k-1}$ if $k - 1 > 0$ or 0 otherwise.
- Allocated Premium: $0.65P$.

- B/O Spread: $(B/O) = 0.03(0.65P) = 0.0195P$.
- Return: $R_k = (IB_k + 0.65P - 0.0195P)\rho_k = (IB_k + 0.6305P)\rho_k$.
- Gross Year End: $GYE_k = IB_k + 0.65P - 0.0195P + R_k = IB_k + 0.6305P + R_k$.
- Annual Management Charge: $AMC_k = 0.011GYE_k$
- Net Year End: $NYE_k = GYE_k - AMC_k$.

Once we have completed this table (table 2 on page 7), we can continue by modelling the cashflow account for the 20 year term.

4 Projecting the Cashflow Account

To project the cashflow account, that is the account that tracks profit and loss from the insurers perspective, we consider all premiums received, the expected benefit payments as well as other expenses over the 20 year term. For this projection, we assume an annual interest rate $i = 2\%$ p.a, initial expenses of £800, renewal expenses £$0.03P$ and terminal expenses at 2% of the death, redundancy or surrender benefit. We have death benefit in year k defined as $B_k^d = \max\{7000, NYE_k\}$, redundancy benefits defined as $B_k^r = (1 + \frac{k}{200}) \cdot NYE_k$, and surrender benefits $B_k^s = 0.6kP$. With these values, we can calculate the following values:

- Unallocated Premium: $UP = 0.35P$
- B/O Spread: $(B/O) = 0.03(0.65P) = 0.0195P$.
- Expected Expenses: $\mathbb{E}(E_k) = 800\delta_k + 0.03P + 0.02 \cdot (1.02)^{-1} \sum_{i=1}^{3} (aq)_{57+k}^{r_i} \cdot B_k^{r_i}$, where $\delta_k = 1$ if $k = 1$ or 0 otherwise.
- Interest: $I_k = (UP + (B/O) - \mathbb{E}(E_k)) \cdot i$
- Annual Management Charge: $AMC_k = 0.011 \cdot GYE_k$
- Expected Extra Benefit: Once again let decrements r_1, r_2, r_3 refer to decrements d, r, s. The expected extra benefit for decrement r_i is $\mathbb{E}(XB_k^{r_i}) = (aq)_{57+k}^{r_i}(B_k^{r_i} - NYE_k)$
- Cashflow: $CF_k = UP + (B/O) - \mathbb{E}(E_k) + I_k + AMC_k - \sum_{j=1}^{3} \mathbb{E}(XB_k^{r_i})$.
- Expected Present Value of Cashflow: $\text{EPV}(CF_k) = CF_k \cdot (1.05)^{-k} \cdot {}_{k-1}(ap)_{58}$.
  - Where:

$$ {}_{k-1}(ap)_{58} = \begin{cases} 1, & \text{if } k = 1, \\ \prod_{j=1}^{k-1}(1 - (aq)_{57+j}), & \text{otherwise.} \end{cases} \tag{4.1} $$

You can view the account projection in table 3 on page 7. Now we have found the expected present values of the cashflows, we can calculate:

$$ \text{NPV(Profit)} = \sum_{k=1}^{20} \text{EPV}(CF_k) = £5,083.60. $$

To calculate our profit margin we also need to calculate:

$$ \text{NPV(Premiums)} = P\left(\sum_{k=1}^{20} {}_{k-1}(ap)_{58} \cdot (1.05)^{-(k-1)}\right) = £15,151.66. $$

Finally we can calculate:

$$\text{Profit Margin} = \frac{\text{NPV(Profit)}}{\text{NPV(Premiums)}} = \frac{£5,083.60}{£15,151.66} = 33.55\%.$$

With the profit margin in this example at 33.55%, we have shown that even under the new death, redundancy and surrender rates, our contract is expected to remain profitable under the specified assumptions.

5 Changing the Surrender Payment Schedule

We now consider alternative policies in which surrender benefits are paid at the end of the $\frac{1}{m}$th of the year in which surrender occurs, for $m \in \{2, 4, 8, 16, 32, 64\}$, rather than at the end of the policy year. To isolate the effect of surrender payment timing, the unit fund projections, annual management charges and cash account interest calculations are held unchanged, as shown in table 2 and table 3. This is a simplifying assumption, as earlier payments could otherwise affect the accumulation of interest and the associated cashflows.

To account for the earlier payments, we use the claims acceleration approximation. Assuming surrender occurs uniformly throughout each policy year, the average payment time for benefits payable at the end of each $\frac{1}{m}$th of a year is $\frac{m+1}{2m}$ years after the start of the policy year. This is $\frac{m-1}{2m}$ years earlier than an end-of-year payment.

We therefore approximate the accumulated value of the surrender benefit $B_k^s = 0.6kP$ at the end of the policy year by applying the factor $(1.02)^{\frac{m-1}{2m}}$. Since the expected extra surrender benefit $\mathbb{E}(XB_k^s)$ is evaluated at year-end, this accumulation factor reflects the earlier payment without changing the timing of the cashflow projection. As $m \to \infty$, the factor approaches $(1.02)^{1/2}$, corresponding to the claims acceleration approximation for immediate surrender payments.

Using this claims acceleration approximation, we can now calculate our new expected extra surrender benefit:

$$\mathbb{E}(XB_k^s) = (aq)_{57+k} \left((1.02)^{\frac{m-1}{2m}} \cdot B_k^s - NYE_k \right). \tag{5.1}$$

The corresponding expected terminal expenses must also be adjusted. We can redefine our expected expenses as:

$$\mathbb{E}(E_k) = 800\delta_k + 0.03P + 0.02 \cdot (1.02)^{-1} \left((aq)^s_{57+k} \cdot B_k^s \cdot (1.02)^{\frac{m-1}{2m}} + \sum_{i=1}^{2} (aq)^{r_i}_{57+k} \cdot B_k^{r_i} \right), \tag{5.2}$$

where once again:

$$\delta_k = \begin{cases} 1, & \text{if } k = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Now we have these new values, we can do exactly as we did in section 4, and recalculate the expected cashflows using the adjusted surrender benefits and expenses. We can do this for all values of $m = 2, 4, 8, 16, 32, 64$. We can also look at what happens when $m \to \infty$. If we look at the accumulation factor used in our approximation: $(1.02)^{\frac{m-1}{2m}}$, if we take the limit of this:

$$\lim_{m \to \infty} (1.02)^{\frac{m-1}{2m}} = (1.02)^{\frac{1}{2}}.$$

We see that if we were to pay surrender benefits immediately on surrender, we need to adjust our expected extra surrender benefit as well as our expected expenses as:

$$\mathbb{E}(XB_k^s) = (aq)_{57+k}\left((1.02)^{\frac{1}{2}} \cdot B_k^s - NYE_k\right),$$

$$\mathbb{E}(E_k) = 800\delta_k + 0.03P + 0.02 \cdot (1.02)^{-1}\left((aq)_{57+k}^s \cdot B_k^s \cdot (1.02)^{\frac{1}{2}} + \sum_{i=1}^{2}(aq)_{57+k}^{r_i} \cdot B_k^{r_i}\right).$$

Once we implement these new measures, we get the following results:

m	NPV(Profit)
1	£5,083.60
2	£5,070.51
4	£5,063.94
8	£5,060.65
16	£5,059.01
32	£5,058.18
64	£5,057.77
∞	£5,057.36

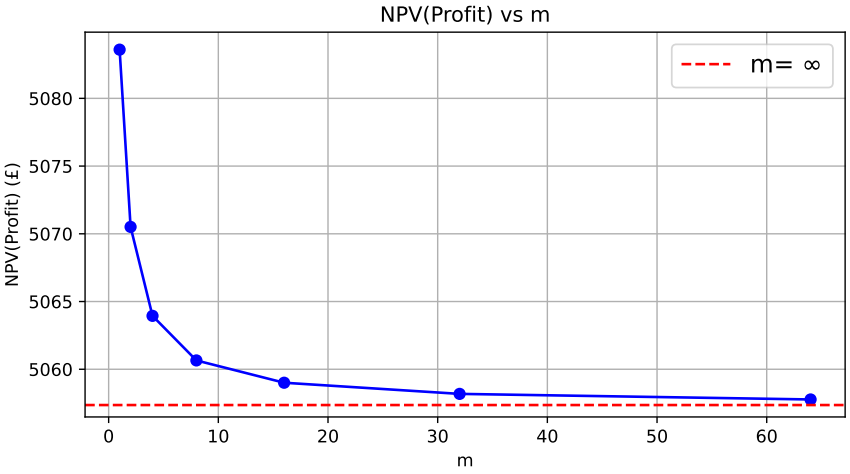


Figure 1: Table and graph showing how NPV(Profit) varies with the time of surrender benefit payment, where payments occur at $\frac{1}{m}$th of a year.

From the graph in fig. 1, we can see that our NPV(Profit) decreases as m increases, and appears to converge to the $m = \infty$ line. This trend can be directly explained from the formula for the expected extra surrender benefit in eq. (5.1). As m increases, the exponent $\frac{m-1}{2m}$ increases, approaching $\frac{1}{2}$ and thus the value $(1.02)^{\frac{m-1}{2m}}$ increases, leading to less discounting on B_k^s. This makes the present value of the benefit payment larger and thus the expected extra benefit payment is larger. This can also be said for the expected expenses in eq. (5.2). Similar to in eq. (5.1), the discount factor makes the present value of the benefit payment larger, and thus the expenses increase as $m \to \infty$. Both of these quantities have an effect on profit. As the cash outflow increases, and premiums stay the same, we have a decreasing profit.

6 Conclusion

This project demonstrates how under the assumptions given, our unit linked policy is expected to remain profitable across a range of scenarios. By first adjusting our multiple decrement table to account for the 8% decrease in death rate, as well as the 5% increase in surrender rate, we obtain the basis for our cashflow projections. We first begin by projecting our unit fund account, which has an annual management charge of 1.1%, a B/O spread of 3% and a time varying return. This allows us to calculate net year end values throughout the 20 year term. We then projected the cashflow account, taking into account our new multiple decrement table. This account had an interest rate of $i = 2\%$, and yielded an expected present value of profit at £5,083.60 when all benefits and expenses are paid annually. This resulted in a healthy profit margin of 33.55%.

Next we explored the sensitivity of profit on the timing of surrender payments. We considered payments at $\frac{1}{m}$th of a year for $m \in \{2, 4, 8, 16, 32, 64\}$. To do this we used an adjustment factor of $(1.02)^{\frac{m-1}{2m}}$ which captures the effect of paying benefits part way through the year. As m increased from 2 to 64, the net present value of profit converged, notably to the case that $m \to \infty$, where the discount factor $\lim_{m\to\infty}(1.02)^{\frac{m-1}{2m}} = (1.02)^{\frac{1}{2}}$ was used. We saw the net present value of profit go from £5,083.60 when $m = 1$ to £5,057.36 when $m \to \infty$. Since the net present values of premiums remained constant, at £15,151.66, we have shown that under the specified assumptions and claims acceleration approximation, the expected present value of profit remains positive across all surrender payment schedules considered.

A Tables

Year	Age	Initial Balance	Allocated Premium	B/O Spread	Return	GYE	AMC	NYE
1	58	£0.00	£1,300.00	£39.00	£31.53	£1,292.53	£14.22	£1,278.31
2	59	£1,278.31	£1,300.00	£39.00	£63.48	£2,602.79	£28.63	£2,574.16
3	60	£2,574.16	£1,300.00	£39.00	£95.88	£3,931.04	£43.24	£3,887.80
4	61	£3,887.80	£1,300.00	£39.00	£128.72	£5,277.52	£58.05	£5,219.46
5	62	£5,219.46	£1,300.00	£39.00	£162.01	£6,642.48	£73.07	£6,569.41
6	63	£6,569.41	£1,300.00	£39.00	£195.76	£8,026.17	£88.29	£7,937.88
7	64	£7,937.88	£1,300.00	£39.00	£229.97	£9,428.85	£103.72	£9,325.14
8	65	£9,325.14	£1,300.00	£39.00	£264.65	£10,850.79	£119.36	£10,731.43
9	66	£10,731.43	£1,300.00	£39.00	£299.81	£12,292.24	£135.21	£12,157.03
10	67	£12,157.03	£1,300.00	£39.00	£335.45	£13,753.48	£151.29	£13,602.19
11	68	£13,602.19	£1,300.00	£39.00	£379.01	£15,242.20	£167.66	£15,074.54
12	69	£15,074.54	£1,300.00	£39.00	£424.89	£16,760.42	£184.36	£16,576.06
13	70	£16,576.06	£1,300.00	£39.00	£473.22	£18,310.28	£201.41	£18,108.87
14	71	£18,108.87	£1,300.00	£39.00	£524.16	£19,894.03	£218.83	£19,675.20
15	72	£19,675.20	£1,300.00	£39.00	£577.88	£21,514.08	£236.65	£21,277.42
16	73	£21,277.42	£1,300.00	£39.00	£634.55	£23,172.97	£254.90	£22,918.07
17	74	£22,918.07	£1,300.00	£39.00	£694.35	£24,873.42	£273.61	£24,599.81
18	75	£24,599.81	£1,300.00	£39.00	£757.50	£26,618.32	£292.80	£26,325.51
19	76	£26,325.51	£1,300.00	£39.00	£824.21	£28,410.72	£312.52	£28,098.21
20	77	£28,098.21	£1,300.00	£39.00	£894.72	£30,253.92	£332.79	£29,921.13

Table 2: Unit fund account projection.

Year	Unallocated Premium	B/O Spread	$\mathbb{E}$(Expenses)	Interest	AMC	$\mathbb{E}(XB_k^d)$	$\mathbb{E}(XB_k^r)$	$\mathbb{E}(XB_k^s)$	CF	EPV(CF)
1	£700.00	£39.00	£861.95	(£2.46)	£14.22	£4.05	£0.03	(£5.73)	(£109.55)	(£104.33)
2	£700.00	£39.00	£63.87	£13.50	£28.63	£3.41	£0.16	(£12.75)	£726.43	£606.72
3	£700.00	£39.00	£65.28	£13.47	£43.24	£2.62	£0.46	(£18.57)	£745.93	£545.70
4	£700.00	£39.00	£66.87	£13.44	£58.05	£1.64	£1.40	(£23.81)	£764.39	£493.58
5	£700.00	£39.00	£68.71	£13.41	£73.07	£0.44	£2.66	(£31.13)	£784.79	£448.29
6	£700.00	£39.00	£70.96	£13.36	£88.29	£0.00	£4.75	(£39.65)	£804.59	£406.25
7	£700.00	£39.00	£74.93	£13.28	£103.72	£0.00	£10.43	(£48.63)	£819.28	£364.49
8	£700.00	£39.00	£77.15	£13.24	£119.36	£0.00	£13.92	(£58.72)	£839.24	£325.11
9	£700.00	£39.00	£79.89	£13.18	£135.21	£0.00	£19.21	(£69.13)	£857.42	£289.23
10	£700.00	£39.00	£82.80	£13.12	£151.29	£0.00	£25.25	(£81.45)	£876.81	£257.02
11	£700.00	£39.00	£86.77	£13.04	£167.66	£0.00	£34.99	(£94.61)	£892.56	£226.85
12	£700.00	£39.00	£88.24	£13.02	£184.36	£0.00	£47.97	(£84.51)	£884.68	£193.90
13	£700.00	£39.00	£89.53	£12.99	£201.41	£0.00	£53.59	(£94.44)	£904.72	£172.02
14	£700.00	£39.00	£86.53	£13.05	£218.83	£0.00	£44.43	(£106.84)	£946.76	£156.80
15	£700.00	£39.00	£85.12	£13.08	£236.65	£0.00	£40.36	(£117.12)	£980.37	£143.51
16	£700.00	£39.00	£85.25	£13.08	£254.90	£0.00	£40.33	(£130.04)	£1,011.44	£132.01
17	£700.00	£39.00	£86.94	£13.04	£273.61	£0.00	£44.90	(£147.25)	£1,041.05	£121.63
18	£700.00	£39.00	£80.24	£13.18	£292.80	£0.00	£23.82	(£140.28)	£1,081.19	£113.08
19	£700.00	£39.00	£71.14	£13.36	£312.52	£0.00	£8.67	(£80.42)	£1,065.48	£101.50
20	£700.00	£39.00	£63.68	£13.51	£332.79	£0.00	£3.84	£0.00	£1,017.78	£90.24

Table 3: Cashflow account projection.